

1. Substitution Method

(a) $T(n) = T\left(\frac{4n}{7}\right) + T\left(\frac{n}{3}\right) + \Theta(n)$

Upper bound: $T(n) \leq cn$ for sufficiently large c .

$$\begin{aligned} T(n) &\leq c \cdot \frac{4n}{7} + c \cdot \frac{n}{3} + \alpha n = cn \left(\frac{4}{7} + \frac{1}{3} \right) + \alpha n = \frac{19}{21} cn + \alpha n \\ &\leq cn \iff \alpha \leq \frac{2c}{21} \iff c \geq \frac{21\alpha}{2} \end{aligned}$$

Base case: $T(c_0) = 1 \leq c \cdot c_0$ for $c \geq 1/c_0$.

Lower bound: $T(n) \geq dn$ for sufficiently small $d > 0$.

$$\begin{aligned} T(n) &\geq d \cdot \frac{4n}{7} + d \cdot \frac{n}{3} + \alpha' n = \frac{19}{21} dn + \alpha' n \\ &\geq dn \iff d \leq \frac{21\alpha'}{2} \end{aligned}$$

$$\boxed{T(n) = \Theta(n)}$$

(b) $T(n) = 2T\left(\frac{3n}{7}\right) + T\left(\frac{n}{7}\right) + \Theta(n)$

Upper bound: $T(n) \leq cn \log n$ for sufficiently large c .

$$\begin{aligned} T(n) &\leq 2c \cdot \frac{3n}{7} \log \frac{3n}{7} + c \cdot \frac{n}{7} \log \frac{n}{7} + \alpha n \\ &= \frac{6cn}{7} (\log n + \log \frac{3}{7}) + \frac{cn}{7} (\log n + \log \frac{1}{7}) + \alpha n \\ &= cn \log n \underbrace{\left(\frac{6}{7} + \frac{1}{7} \right)}_{=1} + cn \underbrace{\left(\frac{6 \log(3/7) + \log(1/7)}{7} \right)}_{=: \beta < 0} + \alpha n \\ &= cn \log n + (\beta c + \alpha) n \\ &\leq cn \log n \iff \beta c + \alpha \leq 0 \iff c \geq \alpha/|\beta| \end{aligned}$$

Base case: n_0 large enough so that $cn_0 \log n_0 \geq T(n_0)$.

Lower bound: $T(n) \geq dn \log n$. With the same approach as before:

$$T(n) \geq dn \log n + (\beta d + \alpha') n \geq dn \log n \iff d \leq \alpha'/|\beta|$$

$$\boxed{T(n) = \Theta(n \log n)}$$

2. Master Method

(a) $T(n) = 9T\left(\frac{n}{3}\right) + n^2$

$$\begin{aligned} a = 9, \quad b = 3, \quad \log_b a = 2 &\implies n^{\log_b a} = n^2 \\ f(n) = n^2 = \Theta(n^2 \log^0 n) &\implies \text{Case 2 } (k = 0) \end{aligned}$$

$$\boxed{T(n) = \Theta(n^2 \log n)}$$

(b) $T(n) = 2T\left(\frac{2n}{3}\right) + n^{4/3}$

$$\begin{aligned} a = 2, \quad b = \frac{3}{2}, \quad \log_{3/2} 2 = \frac{\ln 2}{\ln 3 - \ln 2} \approx 1.71 \\ f(n) = n^{4/3} \implies \frac{4}{3} < \log_{3/2} 2 \implies \text{Case 1} \end{aligned}$$

$$\boxed{T(n) = \Theta\left(n^{\log_{3/2} 2}\right)}$$

(c) $T(n) = 27T\left(\frac{n}{3}\right) + n^3 \log n$

$$\begin{aligned} a = 27, \quad b = 3, \quad \log_b a = 3 &\implies n^{\log_b a} = n^3 \\ f(n) = n^3 \log n = \Theta(n^3 \log^1 n) &\implies \text{Case 2 } (k = 1) \end{aligned}$$

$$\boxed{T(n) = \Theta(n^3 \log^2 n)}$$

(d) $T(n) = 5T\left(\frac{n}{2}\right) + n^2 \log \log n$

$$\begin{aligned} a = 5, \quad b = 2, \quad \log_2 a = \log_2 5 \approx 2.32 \\ f(n) = n^2 \log \log n \implies 2 < \log_2 5 \implies \text{Case 1} \end{aligned}$$

$$\boxed{T(n) = \Theta\left(n^{\log_2 5}\right)}$$

3. Recursion Tree Method

(a) $T(n) = 2T(n-3) + O(1)$

(i) **Height:** $n \rightarrow n-3 \rightarrow n-6 \rightarrow \dots \rightarrow O(1)$, so $h = \lfloor n/3 \rfloor$.

(ii) **Work per level:** Level k has 2^k nodes, each $O(1) \implies O(2^k)$.

(iii) **Total:**

$$T(n) = \sum_{k=0}^{n/3} O(2^k) = O(2^{n/3+1} - 1)$$

Leaves: $2^{n/3}$ nodes each contributing $T(c) = 1$, so $T(n) = \Omega(2^{n/3})$.

$$\boxed{T(n) = \Theta(2^{n/3})}$$

(b) $T(n) = 3T\left(\frac{n}{3}\right) + \frac{n}{\log n}$

(i) **Height:** $n \rightarrow n/3 \rightarrow n/9 \rightarrow \dots \rightarrow 1$, so $h = \log_3 n$.

(ii) **Work per level:** Level k has 3^k nodes of size $n/3^k$:

$$3^k \cdot \frac{n/3^k}{\log(n/3^k)} = \frac{n}{\log n - k \log 3}$$

(iii) **Total:**

$$T(n) = \sum_{k=0}^{\log_3 n - 1} \frac{n}{\log n - k \log 3}$$

Substituting $j = \log_3 n - k$:

$$= \sum_{j=1}^{\log_3 n} \frac{n}{j \log 3} = \frac{n}{\log 3} \underbrace{\sum_{j=1}^{\log_3 n} \frac{1}{j}}_{\sum_{j=1}^m \frac{1}{j} = \Theta(\log m)} = \Theta\left(\frac{n}{\log 3} \cdot \log \log_3 n\right) = \Theta(n \log \log n)$$

$$\boxed{T(n) = \Theta(n \log \log n)}$$